

ERRATUM TO THE 2 SEPTEMBER PUBLIC DRAFT

The 2 September public draft reported $D_3(v_T) = -59072$. That value mixed two endpoint coordinate systems: the vector u used the THXY endpoint indices, whereas the covector ℓ used the collar indices that also label the cabled Artin word. The two tables order the m_2 wickets in opposite directions. In the collar table, the same physical cup has endpoints 2 and 87, so

$$u = e_2 - e_{87}, \quad \ell = e_{87}^* - e_2^*.$$

With both objects in that one table, the exact calculation gives $[\varepsilon^3]\ell(\rho(W) - I)u = -328$ and hence $D_3(v_T) = (-2)^3(-328) = 2624$. The nonvanishing conclusion is unchanged. Using the THXY coordinates consistently gives -2496 ; this is a coordinate check, not a second geometric class. The Lean literals, public receipt, Appendix A, and Remark 12.5 have been updated together. The value 2624 belongs to the currently frozen data. If the unresolved framing record in Assumption A1 changes the collar word, the numerical value must be recomputed. The superseded public PDF had SHA-256 prefix 8E621FD9.

AN EXPLICIT DEGREE-494 CANDIDATE CLASS FOR THE CAPPELL–SHANESON SPHERE $X(41, 189, 73)$

ABSTRACT. We describe a candidate obstruction to the standardness of the Cappell–Shaneson homotopy sphere associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

The finite part of the calculation is exact. A 45,360-letter cabled Artin word acts through the unreduced Burau representation on an 88-dimensional module over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$. A single detector gives

$$D_3(v_T) = 2624, \quad D_3(\xi) = 0,$$

and a separate grading calculation places the proposed class in quantum degree 494. We give all conventions, the generator matrices, the cabling map, immutable input hashes, an independently executable reconstruction, and the complete degree-three vector.

The passage from this finite calculation to a closed four-manifold invariant is conditional. We restate every external theorem with its complete hypotheses and collect the unresolved candidate-specific geometry into five numbered assumptions. Under those assumptions the detector annihilates the one-, two-, and three-handle relations, so the degree-494 class survives in the rational $\mathfrak{sl}_2$ skein lasagna module. Since the corresponding graded piece of the standard 4-sphere vanishes, the assumptions imply that the candidate is not diffeomorphic to S^4 . The accompanying Lean development verifies the finite arithmetic and the abstract quotient implication, but deliberately does not instantiate the geometric assumptions. Thus this paper is a self-contained account of a conditional candidate proof and a precise list of what an independent review must still establish.

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1. INTRODUCTION

The smooth four-dimensional Poincaré conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4 . The Cappell–Shaneson construction supplies a concrete family of homotopy 4-spheres from mapping tori of automorphisms of T^3 . Many members of this family have been

proved standard, but the construction remains a useful test bed because the handle description is explicit enough to support exact calculation.

This article studies the member denoted $X(41, 189, 73)$. The notation refers to a standard-form Cappell–Shaneson matrix; the actual matrix is displayed in [equation \(2\)](#). The proposed obstruction uses the $\mathfrak{sl}_2$ skein lasagna module of Morrison–Walker–Wedrich and its handle-decomposition formulas. Its logic has three steps.

- (1) Construct a class in the one-handle presentation and evaluate a divided cubic detector on it. The value is the nonzero integer 2624 and the class has quantum degree 494.
- (2) Prove, conditional on the stated geometric identifications, that the same detector kills every two-handle relation and the six relations from a chosen system of three attaching spheres. Linear algebra then makes the detector descend to the complete quotient.
- (3) Use the published four-handle and standard-sphere statements to compare the resulting degree-494 class with the degree-494 piece for S^4 .

The first step contains a large but elementary exact computation. The second step contains the principal geometric risk. The third step is short once its coefficient and grading hypotheses are stated correctly. We do not blur these three levels. In particular, a JSON receipt or a Lean structure field is never offered as a proof of a geometric assertion.

1.1. Status of the main statement. The main result of this paper is the conditional implication in [theorem 1.1](#). It is not an unconditional disproof of the smooth four-dimensional Poincaré conjecture. The exact calculations can be reproduced from the public repository, while the assumptions marked “candidate-specific; not certified” require independent geometric proof. This division is part of the mathematical statement, not a disclaimer added after it.

Theorem 1.1 (Conditional candidate obstruction). *Assume A1–A5 below. Then the rational $\mathfrak{sl}_2$ skein lasagna module of $X(41, 189, 73)$ contains a nonzero class in bidegree $(0, 494)$. Consequently*

$$X(41, 189, 73) \not\cong_{\text{diff}} S^4.$$

Together with the Cappell–Shaneson homotopy-sphere criterion, the same assumptions imply a counterexample to the smooth four-dimensional Poincaré conjecture.

The proof appears in [section 12](#). Most of the paper is devoted to making the words “assume A1–A5” completely explicit.

2. ASSUMPTION REGISTER AND LOGICAL ARCHITECTURE

Every load-bearing statement not proved in this paper and not obtained by an exact finite computation appears here. The status labels have the following fixed meanings.

- *External theorem restatement*: a published result is used with no hypothesis removed and no conclusion strengthened. The exact source number and scope are restated in [section 3](#).
- *Candidate-specific; certified by exact computation*: the statement is about immutable finite input and can be checked by a listed program using exact integer or rational arithmetic. The program does not certify any extra geometric interpretation.
- *Candidate-specific; not certified*: the statement identifies a finite model, movie, collar, or surface with the actual geometric object used by a published theorem. This is a mathematical premise still needing proof or independent review.

2.1. Published inputs and exact finite facts. The proof invokes published results E1–E10, each restated with full hypotheses in [section 3](#): the MWW one-, two-, three-, and four-handle formulas; the BHPW strict functoriality theorem; the BPW quantum-trace universal property; the Horvat–Jabłonowski attaching-sphere theorem; the B^4 evaluation and smooth invariance of the skein lasagna module; and the Cappell–Shaneson homotopy-sphere criterion. These are citations, not new hypotheses invented for the candidate. The alternative van Kampen, Wang–Mayer–Vietoris, Poincaré-duality, Hurewicz, and Whitehead route is recorded separately as E11–E14.

Two finite data statements are proved by exact replay rather than assumed.

Certified finite fact C1 (Cut and point-push input). The immutable planar-diagram input has SHA-256 E6912A64457557469E5C691B4D57ABDBF4C45ADB05492777C574223D0C06F8A. Its exact traversal has 2,126,291 crossings, 88 cross-cut connectors, 227 split circle factors, and the registered oriented point-push chronology. The reconstruction and the independent checks are given in [sections 5 and 8](#).

Certified finite fact C2 (Chosen-sphere finite model). The three public PL data sets describe connected, oriented, genus-zero combinatorial surface models. Their class columns are

$$(-1311, 8608, -1)^T, \quad (-189, 1241, 0)^T, \quad (41, -269, 1)^T,$$

and the determinant of the resulting 3×3 matrix is 1. The row counts, Euler calculations, and determinant are recomputed in [section 11](#). The checker performs all-row PL self-intersection and slab-gap tests only for the comb bands. It has no coordinates for the finger annuli, bigon, or cap, and therefore does not certify pairwise-disjoint embedded surfaces. That geometric statement remains part of Assumption [A4](#).

2.2. The five outstanding assumptions.

Assumption A1 (Actual presentation and legal collar). The frozen framed diagram is the actual reduced handle presentation of the Cappell–Shaneson surgery manifold associated with [equation \(2\)](#). Its attaching-component labels, product framings, normal fields, basepoints, registered point-push chronology, collar, and detector insertion disk are carried through one common framed movie. The chronology is a legal linear extension of that movie; coordinate changes transport the operator, vector, shadow, covector, collar, basepoint, and insertion point simultaneously. The finite data of C1 are the finite model of this same geometric object, and the Cappell–Shaneson criterion E10 applies to its surgery description.

Status: candidate-specific; not certified.

Assumption A2 (Coefficient, endpoint, filtration, and grading binding). The hypotheses of E1, E8, and E9 hold for the actual cut coefficient. The paired annuli give the action-compatible Hattori class $\eta_R[T_1]$, whose endpoint shadow is $u = e_2 - e_{87}$ in the module of [definition 7.1](#), with u and ℓ read from the same collar position table. The registered word is the actual point-push word and lies in the third pure-braid filtration term on the whole endpoint module. Resealing the class gives $s_0 = (0, e_{m_2} + e_{r_{xy}})$, and the four geometric grading contributions are $-44, +227, +315, -4$.

Status: candidate-specific; not certified.

Assumption A3 (All-state two-handle cubic cocone). The MWW two-handle formula E2 applies to the actual framed attaching link and relative class. For every finite cable state, the statewise divided-cubic rows use one common physical-copy target, one sign convention, and the actual core counts. They satisfy the beta and psi equations [equations \(69\) and \(70\)](#), and the top-cell projection is compatible with those actual maps after the genuine quantum-trace shadow.

Status: candidate-specific; not certified.

Assumption A4 (Actual chosen spheres and their maps). The finite surfaces of C2 are embedded with the stated framings and class coordinates in one actual post-two-handle boundary, are pairwise disjoint, and meet every hypothesis of the Horvat–Jabłonowski result E7. The MWW three-handle results E3 and E4 apply to these spheres. On the whole source, their actual undotted and once-dotted maps have the leading forms $\text{id}_{\text{old}} \otimes U_j$ and $\text{id}_{\text{old}} \otimes D_j$ after the fixed coordinate transports, and all positive-order transport corrections enter only in order at least h^4 after multiplication by the cubic anomaly.

Status: candidate-specific; not certified.

Assumption A5 (Rational graded closure and standard-sphere comparison). The four-handle isomorphism E5 has bidegree $(0, 0)$ in the convention of the selected class. The integral B^4 and S^4 computations E5 and E6 extend to the rational coefficient convention so that

$$\mathcal{S}_{0,q}^2(S^4; \mathbb{Q}) = 0 \quad (q \neq 0).$$

The smooth-invariance isomorphism preserves that quantum grading.

Status: candidate-specific; not certified.

assumption	single question for independent review
A1	Is the based, framed movie—including collar and insertion point—the actual Cappell–Shaneson presentation?
A2	Does that actual coefficient produce the stated Hattori class, endpoint vector, whole-module filtration, raw state, and grading?
A3	Do the actual beta/psi maps form the all-state divided-cubic cocone?
A4	Are the finite surfaces the actual attaching-sphere basis, and do their whole-source maps have the stated leading Frobenius form?
A5	Do rational coefficients and the grading convention carry the class unchanged through the four-handle and S^4 comparison?

3. PUBLISHED RESULTS USED, WITH THEIR COMPLETE SCOPE

This section records the mathematical content imported from the literature. The wording is a faithful restatement rather than a quotation. In each case the hypotheses are at least those in the cited source, and the conclusion is not strengthened. Candidate-specific identifications appear only in the separate assumptions of [section 2](#).

3.1. One- and two-handle formulas.

External Result E1 (MWW Theorem 4.7). Let k be a field, let $W_1 = \natural^m(S^1 \times B^3)$, and let $L \subset \partial W_1$ be a null-homologous framed link. Suppose that L meets the belt sphere of the i th one-handle transversely in $2p_i$ points. Cutting along these belt spheres gives a tangle R in the complement of paired balls in S^3 . Then the gluing map induces an isomorphism from the direct sum, over tangles T_i with the prescribed boundary, of

$$\mathrm{KhR}_N \left(R \cup \bigsqcup_i (T_i \sqcup \overline{T}_i); k \right) \left\{ \left(\sum_i p_i \right) (N-1) \right\}$$

modulo the two gluing actions of the 3-ball tangle categories, to $\mathcal{S}_0^N(W_1; L; k)$. The source states that a global grading shift is omitted from this displayed presentation.

Source and scope. This is External Result E1, reproduced with the field, null-homology, transversality, boundary, and grading-shift hypotheses that appear in MWW Theorem 4.7 [9, Theorem 4.7]. Its candidate-specific application is part of Assumption [A2](#); no such application is added to the external statement itself.

External Result E2 (MWW Definition 3.1 and Theorem 3.2). Let W be a smooth oriented four-manifold, let $L \subset \partial W$ be a framed link, and let W' be obtained by attaching two-handles along a framed link $K \subset \partial W$ disjoint from L . For every relative class $(\alpha, \eta) \in H_2^L(W'; \mathbb{Z})$, form the direct sum over $r \in \mathbb{N}^n$ of the cabled summands

$$\mathcal{S}_0^N(W; K(r - \alpha^-, r + \alpha^+) \cup L, \eta^r) \{(1 - N)(2|r| + |\alpha|)\}.$$

Quotient by all signed-colour-preserving braid relations $\beta_i(b)v \sim v$ and all stabilization relations $\psi_i^{[d]}(v) \sim 0$ for $d < N - 1$ and $\psi_i^{[N-1]}(v) \sim v$. Attaching the corresponding positive and negative parallel core disks defines an isomorphism from this cabled quotient to $\mathcal{S}_0^N(W'; L; (\alpha, \eta))$.

Source and scope. This is exactly the scope of MWW Definition 3.1 and Theorem 3.2 [9, Definition 3.1 and Theorem 3.2]. In particular, the anti-parallel braid relations are retained; we do not replace their braid groups by symmetric groups.

3.2. Three- and four-handle formulas.

External Result E3 (MWW Theorem 3.7). Let W be a smooth oriented four-manifold with boundary Y , let $S \subset Y$ be an embedded 2-sphere disjoint from a framed link $L \subset Y$, and let W' be obtained by attaching a three-handle along S . Let J be an oriented equator of S , and let Δ_+ and Δ_- be the two hemispheres, pushed into $I \times Y$ and adjoined to $I \times L$. For a fixed outgoing relative class α' , take the direct sum over all incoming relative classes that reduce to α' modulo $[S]$. Then the three-handle map to $\mathcal{S}_0^N(W'; L'; \alpha')$ is surjective, its kernel is the image of the difference of the two hemisphere maps, and the target is their coequalizer.

Source and scope. This is the precise content of MWW Theorem 3.7 [9, Theorem 3.7]. The direct sum over relative classes is part of the theorem and is not suppressed here.

External Result E4 (MWW Theorem 3.10). Suppose $W_1 \subset W_2 \subset W_3 \subset W_4$ is a fixed handle decomposition in which $W_1 = \natural^m(S^1 \times B^3)$, W_2 is obtained by attaching two-handles along K , W_3 by attaching three-handles along $S_1, \dots, S_p$, and W_4 by attaching four-handles. Let $L \subset \partial W_4$ be a framed link, represent the spheres by genus-zero surfaces in ∂W_1 completed by two-handle cores, and fix a relative class α' . Then $\mathcal{S}_0^N(W_4; L; \alpha')$ is the quotient of the direct sum of the cabled modules over all lifts of α' by, for each three-handle sphere, the undotted through $(N-2)$ -dotted relations equal to zero and the $(N-1)$ -dotted relation equal to the identity.

Source and scope. This is MWW Theorem 3.10 with its handle-decomposition, surface-realization, link, and relative-class hypotheses [9, Theorem 3.10]. For $N = 2$ it yields one undotted relation and one once-dotted-minus-identity relation for each sphere.

External Result E5 (MWW Proposition 3.4 and Corollary 3.5). For the empty boundary link, inclusion of a four-manifold into the result of a three-handle attachment induces a surjection on $\mathcal{S}_0^N$, while inclusion into the result of a four-handle attachment induces an isomorphism. Over the integral coefficient convention of the source, $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$ and is concentrated in bidegree $(0, 0)$.

Source and scope. This is MWW Proposition 3.4 and Corollary 3.5 [9, Proposition 3.4 and Corollary 3.5]. The source uses the two-handlebody identification of Manolescu–Neithalath [8, Proposition 2.1] in the handle formula. The source corollary is integral. Its direct rational use is therefore isolated as Assumption A5 rather than being strengthened here.

External Result E6 (Morrison–Walker–Wedrich Definition 5.4 and Example 5.6). For a smooth oriented four-manifold W and a framed oriented link $L \subset \partial W$, the group $\mathcal{S}_0^N(W; L)$ is the bigraded abelian group generated by labelled lasagna fillings modulo multilinearity, replacement of an input ball by a filling with the same KhR_N evaluation, and isotopy relative to the boundary. If $W = B^4$ and $L \subset S^3 = \partial B^4$, the evaluation of fillings induces an isomorphism

$$\mathcal{S}_0^N(B^4; L) \xrightarrow{\cong} \text{KhR}_N(S^3, L). \quad (1)$$

Source and scope. This is the invariant definition and 4-ball evaluation in Morrison–Walker–Wedrich [10, Definition 5.4 and Example 5.6]. The source states the integral bigraded group. Passage to the precise rational convention and its compatibility with the later handle maps remains visible in Assumption A5.

3.3. Replacement of the three attaching spheres.

External Result E7 (Horvat–Jabłonowski Theorem 5.3). Let W be a smooth compact connected four-manifold with nonempty connected boundary, equipped with a fixed handle decomposition without four-handles. Suppose the decomposition has k three-handles and

$$\partial W_2 \cong M' \# (\#_k(S^1 \times S^2)),$$

where M' is irreducible. For pairwise-disjoint smoothly embedded 2-spheres $\Sigma_1, \dots, \Sigma_k \subset \partial W_2$, the following are equivalent: they may be isotoped, up to permutation and three–three handle slides, to the actual attaching spheres; their exterior is connected; and their homology classes form a $\mathbb{Z}$ -basis of $H_2^{\text{sph}}(\partial W_2)$. When these conditions hold, simultaneous surgery on the spheres produces a closed 3-manifold diffeomorphic to M' .

Source and scope. This is the full statement of Theorem 5.3 in version 3 of Horvat–Jabłonowski [5, Theorem 5.3]. The boundary decomposition, irreducibility, number of spheres, pairwise disjointness, and fixed-handle-decomposition hypotheses are all retained.

3.4. Trace, functoriality, and the Cappell–Shaneson criterion.

External Result E8 (BPW Proposition 3.20 and Theorem 3.21). For a locally pregraded endobcategory $(\mathcal{C}, \Sigma)$ with left duals, the quantum horizontal trace is universal among Σ -twisted quantum preshadows; if right duals also exist, it is a quantum shadow. On the 2-category of locally pregraded endobcategories with duals and degree-preserving structure maps, the quantum horizontal trace is a strict 2-functor.

Source and scope. This combines, without enlarging either conclusion, BPW Proposition 3.20 and Theorem 3.21 [3, Proposition 3.20 and Theorem 3.21].

External Result E9 (BHPW Theorem 4.6). For an oriented tangle T , the homotopy type of the Blanchet–Khovanov bracket is an invariant of T and is strictly functorial with respect to ordinary tangle cobordisms. Under the stated BHPW equivalence from the foam category to the oriented Bar–Natan category, its image is isomorphic to the ordinary Khovanov bracket.

Source and scope. This is the strictification result in BHPW [2, Theorem 4.6]. The immediately following remark in the source labels the extension to knotted webs and four-dimensional singular foams as conjectural. Those objects are excluded from Assumption A2.

External Result E10 (Iwaki Proposition 2.1). Let $A \in \mathrm{SL}(3, \mathbb{Z})$ and let Σ_A^ϵ be obtained from the mapping torus of the induced automorphism of T^3 by surgery on the canonical section circle with either of the two framing parities. Then Σ_A^ϵ is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$.

Source and scope. This is Iwaki Proposition 2.1, which in turn cites the original Cappell–Shaneson construction [6, Proposition 2.1]. The premise that our frozen handle diagram is this surgery manifold is not part of that theorem; it remains Assumption A1.

3.5. Standard topology results represented in Lean. The next four items are not needed once Assumption A1 is used, but they state the longer topology route represented in `T73CSTopology.lean`. They are included because a reader auditing the Lean fields must know the exact external content behind those fields.

External Result E11 (Seifert–van Kampen theorem). Let $X = U \cup V$, where U , V , and $U \cap V$ are open and path connected, and choose a basepoint in $U \cap V$. Then the diagram induced by inclusion

$$\pi_1(U \cap V) \longrightarrow \pi_1(U), \quad \pi_1(U \cap V) \longrightarrow \pi_1(V)$$

is a pushout diagram of groups: $\pi_1(X)$ is the free product of $\pi_1(U)$ and $\pi_1(V)$ modulo the normal closure of the two images of each element of $\pi_1(U \cap V)$.

Source and scope. This is the standard based, path-connected form of van Kampen [4, Section 1.2]. Applying it to the Cappell–Shaneson surgery still requires the geometric identification of the cover and of the map $A - I$; Lean stores that application as a field.

External Result E12 (Wang and Mayer–Vietoris exact sequences). For a fibration $F \rightarrow E \rightarrow S^1$ with monodromy $f : F \rightarrow F$, there is a long exact Wang sequence whose relevant portion is

$$\cdots \rightarrow H_k(F) \xrightarrow{I-f_*} H_k(F) \rightarrow H_k(E) \rightarrow H_{k-1}(F) \xrightarrow{I-f_*} H_{k-1}(F) \rightarrow \cdots$$

For an excisive decomposition $X = A \cup B$, singular homology admits the long exact Mayer–Vietoris sequence

$$\cdots \rightarrow H_k(A \cap B) \rightarrow H_k(A) \oplus H_k(B) \rightarrow H_k(X) \rightarrow H_{k-1}(A \cap B) \rightarrow \cdots,$$

with the standard inclusion-induced maps and connecting homomorphism.

Source and scope. These are the ordinary singular-homology exact sequences [4, Sections 2.2 and 4.2]. The candidate-specific claim that their maps are the matrices in equations (5) and (10) is not part of the general theorem.

External Result E13 (Integral Poincaré duality). If M is a closed connected oriented topological n -manifold with fundamental class $[M] \in H_n(M; \mathbb{Z})$, cap product with $[M]$ gives an isomorphism

$$H^k(M; \mathbb{Z}) \xrightarrow{\cong} H_{n-k}(M; \mathbb{Z})$$

for every k .

Source and scope. This is the integral oriented form of Poincaré duality [4, Section 3.3]. Closedness, connectedness, orientation, and the use of integral coefficients are retained.

External Result E14 (Hurewicz and Whitehead promotion). If a path-connected space X is $(n-1)$ -connected for $n \geq 2$, then $H_i(X; \mathbb{Z}) = 0$ for $i < n$ and the Hurewicz map $\pi_n(X) \rightarrow H_n(X; \mathbb{Z})$ is an isomorphism. If a map between simply connected CW complexes induces an isomorphism on all integral homology groups, then it is a homotopy equivalence.

Source and scope. The first sentence is the Hurewicz theorem and the second is the homological form of Whitehead's theorem for simply connected CW complexes [4, Sections 4.1–4.2]. Applied successively, they show that a simply connected CW complex with the integral homology of S^4 is homotopy equivalent to S^4 . The premise that the candidate has that topology is external to the abstract theorems.

Remark 3.1 (Why the short route is used). The final proof uses Iwaki Proposition 2.1 rather than silently rebuilding the Cappell–Shaneson calculation from these four results. The longer route and its Lean fields remain in [table 1](#); the displayed statements make clear that Lean does not prove their candidate-specific hypotheses.

4. THE CAPPELL–SHANESON INPUT

4.1. The matrix and its elementary invariants.

Definition 4.1. Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix} \in M_3(\mathbb{Z}). \quad (2)$$

The symbol $X(41, 189, 73)$ denotes the framed surgery presentation that is asserted in Assumption [A1](#) to be the Cappell–Shaneson surgery manifold Σ_A^ϵ . The parameter 73 belongs to the historical standard-form naming convention; it is not an entry of [equation \(2\)](#).

Exact computation 4.2 (The two determinant tests). The matrix A has determinant 1, and so does $A - I$.

Proof. Expansion along the first column of A gives

$$\det A = 269 \cdot 189 - 1240 \cdot 41 \quad (3)$$

$$= 50841 - 50840 = 1. \quad (4)$$

Moreover

$$A - I = \begin{pmatrix} -1 & 269 & 1240 \\ 0 & 40 & 189 \\ 1 & 0 & 31 \end{pmatrix}. \quad (5)$$

Expanding along its first row gives

$$\det(A - I) = (-1)(40 \cdot 31) - 269(0 \cdot 31 - 189 \cdot 1) + 1240(0 \cdot 0 - 40 \cdot 1) \quad (6)$$

$$= -1240 + 50841 - 49600 = 1. \quad (7)$$

Every operation in [equations \(3\)](#) and [\(6\)](#) is integer arithmetic. The same identities are checked in Lean by `T73.detA_eq_one` and `T73.detAMinusI_eq_one`. $\square$

Exact computation 4.3 (Integral inverses). The endomorphisms $A - I$ on $H_1(T^3; \mathbb{Z})$ and $(A^{-1})^T - I$ on $H_2(T^3; \mathbb{Z})$ are integral automorphisms.

Proof. For $B = A - I$, define

$$B_{\mathbb{Z}}^{-1} = \begin{pmatrix} 1240 & -8339 & 1241 \\ 189 & -1271 & 189 \\ -40 & 269 & -40 \end{pmatrix}. \quad (8)$$

Multiplying [equation \(5\)](#) by [equation \(8\)](#) on either side gives the identity matrix. For example, the first row of $BB_{\mathbb{Z}}^{-1}$ is

$$\begin{aligned} & (-1240 + 269 \cdot 189 - 1240 \cdot 40, \\ & 8339 - 269 \cdot 1271 + 1240 \cdot 269, \\ & -1241 + 269 \cdot 189 - 1240 \cdot 40) = (1, 0, 0), \end{aligned} \quad (9)$$

and the other six entries are checked by the same integer multiplication.

Since [equation \(4\)](#) gives $A^{-1} \in M_3(\mathbb{Z})$, direct inversion gives

$$C = (A^{-1})^T - I = \begin{pmatrix} 1311 & 189 & -41 \\ -8608 & -1241 & 269 \\ 1 & 0 & -1 \end{pmatrix}. \quad (10)$$

An integral inverse is

$$C_{\mathbb{Z}}^{-1} = \begin{pmatrix} -1241 & -189 & 40 \\ 8339 & 1270 & -269 \\ -1241 & -189 & 39 \end{pmatrix}. \quad (11)$$

Again $CC_{\mathbb{Z}}^{-1} = C_{\mathbb{Z}}^{-1}C = I_3$. These are the matrices used by `T73CSAlgebra.lean`; the Lean proofs expand all nine entries and close them by ring normalization. $\square$

Remark 4.4 (Long topology route). The two inverses have the expected topological consequences once the mapping torus and surgery are identified. Surjectivity of $A - I$ kills the natural coinvariant presentation of the fundamental group after surgery; bijectivity of $A - I$ and $(A^{-1})^T - I$ kills the intermediate homology in the Wang and Mayer–Vietoris sequences. Poincaré duality then gives the homology profile of S^4 . Hurewicz and Whitehead promote simple connectivity plus that homology profile to a homotopy equivalence with S^4 . Those implications are represented by separate fields in `T73CSTopology.lean`; the field types are shown in [table 1](#). This paper uses the shorter published Cappell–Shaneson criterion instead, so no step of this longer route is needed to conceal an extra geometric premise.

Corollary 4.5 (Conditional homotopy-sphere conclusion). *Under Assumption A1 and External Result E10, $X(41, 189, 73)$ is a homotopy 4-sphere.*

Proof. By [equation \(4\)](#), $A \in \mathrm{SL}(3, \mathbb{Z})$. By [equation \(7\)](#), $\det(A - I) = 1$. Assumption A1 identifies the candidate with Σ_A^c ; External Result E10, namely [External Result E10](#), then gives the conclusion. $\square$

4.2. What the topology calculation does not prove. The determinant test proves no smooth standardness statement. It identifies the surgery manifold as a homotopy sphere, conditional on the presentation binding. The proposed obstruction below is needed precisely because the usual homotopy invariants do not distinguish the candidate from S^4 . Conversely, a nonzero finite scalar attached only to the displayed matrix is not automatically a smooth invariant. The handle and naturality assumptions are what connect the scalar to the closed manifold.

5. FROM THE MAPPING TORUS TO THE FROZEN CUT TANGLE

This section explains the geometry that Assumption A1 must bind. Several steps are elementary or finite; the assertion that their composite is the actual framed Cappell–Shaneson handle presentation remains an assumption.

5.1. The mapping-torus relations. Let x_1, x_2, x_3 be the coordinate generators of $\pi_1(T^3) = \mathbb{Z}^3$ and let t denote the base circle of the mapping torus. The mapping-torus presentation has relations

$$m_i = t \phi_A(x_i) t^{-1} x_i^{-1}, \quad i = 1, 2, 3, \quad (12)$$

where ϕ_A is induced by the column action of A . Aitchison–Rubinstein describe each m_i as the boundary of a product annulus assembled from a bottom coordinate arc, its top image, and two vertical connecting arcs [1, pp. 5–9]. The product annulus, rather than a blackboard writhe, specifies the framing used here.

The first column of [equation \(2\)](#) is e_3 , so $\phi_A(x_1) = x_3$. After the section/base-handle cancellation, [equation \(12\)](#) gives

$$m_1 = x_3 x_1^{-1}. \quad (13)$$

Cancelling this complementary one–two handle pair identifies x_1 with x_3 . Under that substitution the three commutator components become

$$[x_1, x_2] \mapsto [x_3, x_2], \quad [x_2, x_3] \mapsto [x_2, x_3], \quad [x_3, x_1] \mapsto [x_3, x_3] = 1. \quad (14)$$

The last component bounds the two nested product bigons obtained by cancelling $x_3 x_3 x_3^{-1} x_3^{-1}$; it is a split zero-framed component in the product model.

Exact computation 5.1 (Reduced word data). The frozen integer word builder applies [equation \(12\)](#) and the two substitutions above to obtain a reduced word m_2 of length 311 with exponent sum $(40, 269)$ in (x_2, x_3) , and a reduced word m_3 of length 1460 with exponent sum $(189, 1271)$.

Proof. The words are generated from the columns $Ae_2 = (269, 41, 0)^T$ and $Ae_3 = (1240, 189, 32)^T$ by the Christoffel-path convention in the public builder, followed by free reduction after $x_1 = x_3$. The checker `scripts/verify_public_geometry_evidence.py` regenerates the paths and their exponent sums from `evidence/public_geometry/cs_presentation.py`. Because the output is an integer word, both the length and the exponent sums are exact. The statement that these paths are the transported top arcs of the actual framed product annuli is not a consequence of word equality and remains in Assumption A1. $\square$

5.2. A relative descending criterion.

Lemma 5.2 (Relative global descending lemma). *Let T be a marked tangle in a 3-ball. Suppose its boundary endpoints and pairing are fixed; its components carry a total order; on each component the branch encountered earlier from the marked initial point is over at every self-crossing; and at every mutual crossing an earlier component is over a later one. Then T is ambient-isotopic relative to the boundary to the boundary-parallel trivial tangle with the same marked endpoint pairing.*

Proof. Place the first component in a thin top layer. Traverse it from the marked initial endpoint. At a self-crossing, the already traversed branch lies over the untraversed branch; at a mutual crossing, the whole component lies over every later component. The traversed arc can therefore be pushed into a boundary collar without crossing the remaining tangle. Continuing produces a bridge disk disjoint from all later components. Remove the first component and repeat. Induction gives disjoint bridge disks for every component, hence an ambient isotopy relative to the boundary. The fixed endpoint pairing determines the unique marked boundary-parallel target. $\square$

Exact computation 5.3 (Descending check for the frozen PD). The exact traversal of the frozen 2,126,291-crossing PD input satisfies the hypotheses of Lemma 5.2. The self-crossing counts for $(r_{xy}, r_{yz}, m_2, m_3, r_{zx})$ are

$$(9, 3, 64631, 1445582, 0), \quad (15)$$

and every mutual crossing follows the strict component order

$$r_{xy} > r_{yz} > m_2 > m_3. \quad (16)$$

Proof. The script `scripts/verify_t73_global_descending.py` reads every crossing, reconstructs the cyclic traversal including the closing connector, checks the over/under condition at every self-crossing, and checks equation (16) at every mutual crossing. It recomputes the counts in equation (15) rather than trusting summary fields. The input identity is the SHA-256 in Certified Fact C1. $\square$

The descending lemma proves that the frozen cut tangle is trivial relative to its marked endpoints. It does not prove that the tangle is the cut of the actual Aitchison–Rubinstein product presentation. Assumption A1 supplies that last whole-link, framing-preserving identification. Laudenbach–Poenaru’s theorem permits a boundary self-diffeomorphism of $\#_4(S^1 \times S^2)$ to extend over the four-dimensional one-handlebody [7]; it does not identify the finite diagram without the preceding geometric construction.

6. COEFFICIENT TRACES AND THE SELECTED HATTORI CLASS

6.1. Coefficient Hochschild homology.

Definition 6.1 (Linear category and bimodule). Let $\mathcal{C}$ be a small $\mathbb{Q}$ -linear category. A coefficient bimodule M assigns a vector space $M(T, T')$ to each ordered pair of objects and has commuting left and right actions

$$\mathcal{C}(S, T) \otimes M(T, T') \longrightarrow M(S, T'), \quad M(T, T') \otimes \mathcal{C}(T', U) \longrightarrow M(T, U). \quad (17)$$

Associativity and unit conditions are the usual ones.

Definition 6.2 (Coefficient trace). The coefficient zeroth Hochschild homology is

$$\mathrm{HH}_0(\mathcal{C}; M) = \left(\bigoplus_{T \in \mathrm{Obj} \mathcal{C}} M(T, T) \right) / \mathrm{span}_{\mathbb{Q}} \{f \cdot m - m \cdot f\}, \quad (18)$$

where $f : S \rightarrow T$ and $m \in M(T, S)$ range over every cyclically composable pair. If $m \in M(T, T)$ is a diagonal cycle, its image in equation (18) is denoted $[m]_T$. When m is the identity-type element obtained from a representable coefficient, we call $[m]_T$ its *Hattori class*.

This definition is deliberately concrete. It is the quotient implemented in `Smooth4PC/CoefficientTrace.lean`. The quotient universal property is ordinary linear algebra: a linear map on the direct sum descends if and only if it kills every vector $f \cdot m - m \cdot f$.

Lemma 6.3 (Descent through the coefficient trace). *Let $r_T : M(T, T) \rightarrow V$ be a family of linear maps satisfying*

$$r_S(f \cdot m) = r_T(m \cdot f) \quad (19)$$

for all cyclically composable f and m . Then there is a unique linear map $\bar{r} : \mathrm{HH}_0(\mathcal{C}; M) \rightarrow V$ whose restriction to $M(T, T)$ is r_T .

Proof. Let r be the sum of the r_T on the algebraic direct sum. By equation (19), $r(f \cdot m - m \cdot f) = 0$ for every generator of the relation space in equation (18). Hence that relation space lies in $\ker r$, so r factors uniquely through the quotient. $\square$

6.2. The Frobenius factor and the diagonal element.

Definition 6.4 (Rank-two Frobenius algebra). Throughout the finite $\mathfrak{sl}_2$ calculation we use

$$\mathbf{A} = \mathbb{Q}[X]/(X^2), \quad \epsilon(1) = 0, \quad \epsilon(X) = 1, \quad (20)$$

with coproduct

$$\Delta(1) = 1 \otimes X + X \otimes 1, \quad \Delta(X) = X \otimes X. \quad (21)$$

The 227 labels below mean the tensor $X^{\otimes 227}$, not the product $X^{227} = 0$ in $\mathbf{A}$.

Definition 6.5 (Selected object and class). Let $U : P_{86} \rightarrow P_{88}$ be the physical oriented one-cup tangle. Let $B_{\mathrm{act}} \in \mathrm{Aut}_{\mathcal{C}}(P_{88})$ be the automorphism in Assumption A2, and set

$$T_1 = B_{\mathrm{act}}^{-1}U. \quad (22)$$

The balanced coefficient equivalence is denoted

$$H_{T, T'} : M_R(T, T') \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(B_{\mathrm{act}}T, B_{\mathrm{act}}T')\{-44\} \otimes \mathbf{A}^{\otimes 227}. \quad (23)$$

Define

$$v_T = H_{T_1, T_1}^{-1}(\mathrm{id}_U \otimes X^{\otimes 227}), \quad \eta_R[T_1] = [v_T]_{T_1} \in \mathrm{HH}_0(\mathcal{C}; M_R). \quad (24)$$

Equation (24) defines the algebraic element once equation (23) is given. The substantive geometric statement is that equation (23) is the coefficient equivalence arising from the actual paired-annulus cut, with compatible left and right actions. That statement is Assumption A2.

For comparison with the earlier relative input, factor $B_{\mathrm{act}} = WF_{\Omega}$ and define

$$T_0 = F_{\Omega}^{-1}U, \quad T_1 = F_{\Omega}^{-1}W^{-1}U. \quad (25)$$

Then associativity, without any commutativity assumption, gives

$$B_{\mathrm{act}}T_0 = WU, \quad B_{\mathrm{act}}T_1 = U. \quad (26)$$

The first equality follows from $WF_{\Omega}F_{\Omega}^{-1}U = WU$; the second follows from $WF_{\Omega}F_{\Omega}^{-1}W^{-1}U = U$.

Definition 6.6 (Relative class). Let $s_{\mathrm{inv}} \in \{\pm 1\}$ be the strict functorial sign assigned to the fixed cancellation movie for WW^{-1} . The relative class is

$$\xi = \eta_R[T_0] - s_{\mathrm{inv}}\eta_R[T_1]. \quad (27)$$

It is not the selected input $\eta_R[T_1]$.

7. THE ENDPOINT MODULE AND THE BURAU ACTION

7.1. The 88-dimensional module.

Definition 7.1 (Endpoint module). Let

$$R_\varepsilon = \mathbb{Z}[\varepsilon]/(\varepsilon^7), \quad t = 1 + \varepsilon, \quad (28)$$

and let

$$E_\varepsilon = R_\varepsilon^{88} = \bigoplus_{i=0}^{87} R_\varepsilon e_i. \quad (29)$$

One may identify this free module with the weight-86 subspace of the 88-fold tensor power of the two-dimensional fundamental module: e_i corresponds to the tensor having the lower-weight basis vector in position i and the upper-weight basis vector in every other position. This explains the dimension $\binom{88}{1} = 88$. The identification of this algebraic model with the actual endpoint shadow is part of Assumption A2.

Definition 7.2 (Unreduced Burau representation). For the Artin generator σ_i of B_{88} , with $1 \leq i \leq 87$, define $\rho_t(\sigma_i)$ to be the identity off the span of e_{i-1}, e_i and to have the block

$$\rho_t(\sigma_i)|_{\langle e_{i-1}, e_i \rangle} = \begin{pmatrix} 1-t & t \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -\varepsilon & 1+\varepsilon \\ 1 & 0 \end{pmatrix}. \quad (30)$$

Its inverse has block

$$\rho_t(\sigma_i^{-1})|_{\langle e_{i-1}, e_i \rangle} = \begin{pmatrix} 0 & 1 \\ t^{-1} & 1-t^{-1} \end{pmatrix}, \quad (31)$$

where

$$t^{-1} = 1 - \varepsilon + \varepsilon^2 - \varepsilon^3 + \varepsilon^4 - \varepsilon^5 + \varepsilon^6 \quad \text{in } R_\varepsilon. \quad (32)$$

For a word $w = \sigma_{i_1}^{s_1} \cdots \sigma_{i_m}^{s_m}$ we use the left action

$$\rho_t(w)v = \rho_t(\sigma_{i_1}^{s_1}) \cdots \rho_t(\sigma_{i_m}^{s_m})v. \quad (33)$$

Thus an implementation acting in place on a column vector must traverse the stored letters from right to left.

Lemma 7.3. *The blocks in equations (30) and (31) are mutual inverses over R_ε .*

Proof. Direct multiplication gives

$$\begin{pmatrix} 1-t & t \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ t^{-1} & 1-t^{-1} \end{pmatrix} = \begin{pmatrix} 1 & (1-t)+t-1 \\ 0 & 1 \end{pmatrix} = I_2. \quad (34)$$

The reverse product is the same calculation. Formula (32) is the geometric-series inverse of $1 + \varepsilon$ modulo ε^7 . $\square$

7.2. From primitive crossings to the cabled word. The public input stores 252 primitive crossing rows, not the expanded 11,340- or 45,360-letter words. Each row records a sign $s \in \{\pm 1\}$, two paired-strand labels $1 \leq i < j \leq 44$, and one of two geometries.

Definition 7.4 (Row expansion). For $i < j$ define the two signed pure-row words

$$L_{ij}^s = \sigma_{j-1}\sigma_{j-2} \cdots \sigma_{i+1}\sigma_i^s\sigma_i^{s-1} \cdots \sigma_{j-2}^{-1}\sigma_{j-1}^{-1}, \quad (35)$$

$$R_{ij}^s = \sigma_i\sigma_{i+1} \cdots \sigma_{j-2}\sigma_{j-1}^s\sigma_{j-1}^{s-1} \cdots \sigma_{i+1}^{-1}\sigma_i^{-1}. \quad (36)$$

The 44-strand word B_{44} is obtained by taking the R rows on segment 6 in increasing horizontal coordinate, followed by the L rows on segment 2 in decreasing horizontal coordinate, and replacing each row by equations (35) and (36).

Definition 7.5 (Two-cabling). The cabling homomorphism on letters is

$$c(\sigma_i) = \sigma_{2i}\sigma_{2i+1}\sigma_{2i-1}\sigma_{2i}, \quad c(\sigma_i^{-1}) = c(\sigma_i)^{-1}. \quad (37)$$

The 88-strand word is $B_{88} = c(B_{44})$.

The use of “homomorphism” in definition 7.5 is algebraic: the code applies equation (37) letter by letter and reverses and changes signs for an inverse letter. The geometric statement that this is the actual oriented cabling of the point-push movie belongs to Assumptions A2 and A1.

Exact computation 7.6 (Word identities). The exact reconstruction gives

$$|B_{44}| = 11340, \quad |B_{88}| = 45360. \quad (38)$$

The canonical JSON SHA-256 values are, respectively,

$$7C2D2F792C2672221A76CAF08A71F560AF0CB7654B4D537C00FAD00B16EFA187$$

and

$$C13D0A3BA9B05F41A6D2C5B4AB12DDECDABEFC1883835891AD3BD2B8955B5FFB.$$

The first forty letters are

$$B_{44}[0:40] = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, \\ 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40), \quad (39)$$

$$B_{88}[0:40] = (2, 3, 1, 2, 4, 5, 3, 4, 6, 7, 5, 6, 8, 9, 7, 8, 10, 11, 9, 10, \\ 12, 13, 11, 12, 14, 15, 13, 14, 16, 17, 15, 16, 18, 19, 17, 18, 20, 21, 19, 20). \quad (40)$$

Proof. The reconstruction is precisely Definitions 7.4 and 7.5. Canonical JSON means UTF-8 encoding with sorted keys where applicable and list separators , and : without spaces. The command in section 15 recomputes the two lengths, the two hashes, and the displayed prefixes from the primitive rows. Neither expanded word is stored in the input. $\square$

8. THE DETECTOR AND THE EXACT CUBIC

8.1. One definition for both inputs. Let $R_h = \mathbb{Q}[h]/(h^7)$ and substitute

$$q = 1 + h, \quad t = q^{-2}, \quad \varepsilon = t - 1. \quad (41)$$

Let Sh_h denote the quantum trace shadow followed by the fixed endpoint identification, let P_{86} be the rational projector onto the through-86 top cell, and let $\widehat{C}(h)$ be the normalized endpoint cap row. These maps are typed on the full endpoint target by Assumptions A2 and A3.

Definition 8.1 (The detector). For a quantum trace class x , define the single series

$$\mathcal{D}_h(x) = \widehat{C}(h)(\rho_h(W) - I)P_{86}\text{Sh}_h(x). \quad (42)$$

Under Assumption A2, the image is contained in h^3R_h . The divided cubic detector on the ordinary coefficient trace is

$$D_3(\bar{x}) = \left. \frac{\mathcal{D}_h(x)}{h^3} \right|_{h=0}, \quad (43)$$

where x is any quantum lift of $\bar{x}$.

Lemma 8.2 (Lift independence). *Formula (43) is independent of the quantum lift.*

Proof. If x' and x have the same specialization at $h = 0$, then $x' - x = hy$. Write $\mathcal{D}_h = h^3\mathcal{D}'_h$. Linearity gives

$$\mathcal{D}'_h(x') - \mathcal{D}'_h(x) = h\mathcal{D}'_h(y), \quad (44)$$

whose reduction modulo h is zero. Thus equation (43) has the same value on x and x' . $\square$

Lemma 8.3 (Specialization of the quantum trace). *Let $q\text{Tr}_q(\mathcal{C}; M)$ be the quotient of $\bigoplus_T M(T, T) \otimes \mathbb{Q}[q, q^{-1}]$ by*

$$L_f(m) - q^{|f|}R_f(m). \quad (45)$$

Then

$$q\text{Tr}_q(\mathcal{C}; M) \otimes_{\mathbb{Q}[q, q^{-1}]} \mathbb{Q}[q, q^{-1}]/(q - 1) \cong \text{HH}_0(\mathcal{C}; M). \quad (46)$$

Proof. Tensor product is right exact. Applying it to the cokernel presentation of $q\text{Tr}_q$ replaces $q^{|f|}$ by 1, so every generator equation (45) becomes $L_f(m) - R_f(m)$. The resulting cokernel is exactly equation (18). No flatness assumption on $q\text{Tr}_q$ is needed. $\square$

Under the endpoint normalization of the public input, the composite row $\widehat{C}(0)P_{86}$ is

$$\ell = e_{87}^* - e_2^*, \quad (47)$$

and the selected shadow vector is

$$u = e_2 - e_{87}. \quad (48)$$

Here $e_i^*(e_j) = \delta_{ij}$. Both formulas use the collar-coordinate table `data/B88_POSITION_TO_PASSAGE_TABLE.json`; its SHA-256 is 119C7E9E74AE6C820DA72A84CDFD5D445D81E6C3AACC209C25D37C323961508. In that table positions 2 and 87 are the negative copy of wicket W_2 and the positive copy of wicket W_{44} , respectively.

8.2. The exact vector calculation.

Exact computation 8.4 (Bureau difference on the selected vector). In E_ε , the first nonzero coefficient of $(\rho_t(B_{88}) - I)u$ occurs in degree 3. Its full coefficient vector is

$$[\varepsilon^3](\rho_t(B_{88}) - I)u = (a_0, \dots, a_{87})^T, \quad (49)$$

where the 88 integers a_i are printed in [appendix A](#). Contracting with [equation \(47\)](#) gives

$$\ell(\rho_t(B_{88}) - I)u = -328\varepsilon^3 + 14596\varepsilon^4 - 410246\varepsilon^5 + 9595271\varepsilon^6. \quad (50)$$

Proof. Start with the sparse column u in [equation \(48\)](#). Traverse the letters of B_{88} from right to left, applying [equation \(30\)](#) for a positive letter and [equation \(31\)](#) for a negative letter. All polynomial products are reduced modulo ε^7 after each operation. Subtract the original column. Finally take coordinate 87 minus coordinate 2 as in [equation \(47\)](#). This deterministic algorithm is implemented in `scripts/recompute_t73_delta3.py`; it uses Python integers only. The input SHA-256 is

0F0F4A568D966367313C29C25CEDDE4FD9ABAD32B9375519E9CDF9168AE0A764.

The program rejects a mismatch in the primitive-row hash, either word hash, the representation block, the cabling rule, or the Hattori word binding. Thus [equation \(50\)](#) is independently reimplementable from the formulas in this paper and the immutable primitive rows. $\square$

Lemma 8.5 (Parameter expansion). In $\mathbb{Z}[h]/(h^7)$,

$$\varepsilon = (1 + h)^{-2} - 1 = -2h + 3h^2 - 4h^3 + 5h^4 - 6h^5 + 7h^6. \quad (51)$$

Proof. The binomial series gives $(1 + h)^{-2} = \sum_{k \geq 0} (-1)^k (k + 1) h^k$. Subtracting 1 and truncating after degree 6 yields [equation \(51\)](#). $\square$

Lemma 8.6 (Commutator order). Let R be an h -adically filtered ring. If $P = I + O(h^a)$ and $Q = I + O(h^b)$ are invertible matrices, then

$$PQP^{-1}Q^{-1} = I + O(h^{a+b}). \quad (52)$$

Consequently, if every generator of a pure-braid subgroup acts as $I + O(h)$, then its r th lower-central subgroup acts as $I + O(h^r)$.

Proof. Write $P = I + h^a P_0$ and $Q = I + h^b Q_0$. Modulo h^{a+b} , the inverse series are $P^{-1} = I - h^a P_0 + O(h^{2a})$ and $Q^{-1} = I - h^b Q_0 + O(h^{2b})$. Multiplication shows that all terms of order strictly below $a + b$ cancel in $PQP^{-1}Q^{-1}$. The lower-central claim follows by induction from $\Gamma_{r+1} = [\Gamma_r, \Gamma_1]$. $\square$

Lemma 8.6 proves the filtration implication. It does not prove that the geometric point-push word belongs to Γ_3 ; that membership and the identification of the word with the actual movie are the unproved parts of Assumption A2.

Proposition 8.7 (Selected value). Under Assumption A2, the detector of [definition 8.1](#) satisfies

$$D_3(\eta_R[T_1]) = 2624 \neq 0. \quad (53)$$

Proof. Assumption A2 and [equation \(26\)](#) identify the leading endpoint shadow of $\eta_R[T_1]$ with u . By [equation \(50\)](#), the coefficients below ε^3 vanish. Hence only the cubic term contributes to the coefficient of h^3 . By [equation \(51\)](#), $[h]\varepsilon = -2$, so

$$[h^3](-328)\varepsilon^3 = (-328)(-2)^3 = 2624. \quad (54)$$

The last integer is nonzero. $\square$

Exact computation 8.8 (The squared shell). Applying the same operator a second time gives

$$\ell(\rho_t(B_{88}) - I)^2 u = -102729600\varepsilon^6 \quad \text{in } R_\varepsilon. \quad (55)$$

Proof. Use the output column of [exact computation 8.4](#) as the input to the same right-to-left word action, subtract that input column, and contract with the same covector [equation \(47\)](#). This is the second calculation performed by the public script. No second detector definition is introduced. $\square$

Exact computation 8.9 (The complete scalar profiles in the h coordinate). Substitution of [equation \(51\)](#) into [equations \(50\)](#) and [\(55\)](#) yields, through degree six,

$$\begin{aligned} \ell(\rho_h(W) - I)u &= 2624h^3 + 221728h^4 + 11760112h^5 \\ &\quad + 520583560h^6, \end{aligned} \quad (56)$$

$$\ell(\rho_h(W) - I)^2 u = -6574694400h^6. \quad (57)$$

Proof. Polynomial substitution in the quotient $\mathbb{Z}[h]/(h^7)$ is finite. For example, the cubic coefficient of the first line is $(-328)(-2)^3 = 2624$. For the second line, $\varepsilon^6 = (-2)^6 h^6 + O(h^7) = 64h^6 + O(h^7)$, and $-102729600 \cdot 64 = -6574694400$. Expanding the remaining powers and collecting coefficients gives [equation \(56\)](#); the public script performs the same convolution with integer lists. $\square$

Proposition 8.10 (The relative input is killed). *With ξ defined by [equation \(27\)](#) and with the same detector [equation \(43\)](#),*

$$D_3(\xi) = 0. \quad (58)$$

Proof. The cancellation sign in [definition 6.6](#) is chosen so that

$$P_{86} \text{Sh}_h(\xi) = (\rho_h(W) - I)u. \quad (59)$$

Substitution of [equation \(59\)](#) into the single definition [equation \(42\)](#) produces the square $\ell(\rho_h(W) - I)^2 u$. By [equation \(55\)](#) it begins in order ε^6 , hence in order h^6 . Its h^3 coefficient is zero. $\square$

Remark 8.11 (Why the two values are not contradictory). The plain shadow coefficient $[h^3]\ell \text{Sh}_h(\xi)$ equals 2624, but that plain shadow is not $D_3(\xi)$. The detector already contains the outer factor $\rho_h(W) - I$. Thus the selected class $\eta_R[T_1]$ receives one factor and has value 2624, whereas ξ already contains one factor and the detector supplies a second, yielding zero in degree three.

9. THE GRADING CALCULATION

Proposition 9.1 (Quantum degree). *Under Assumption A2, the selected raw class has quantum degree 494.*

Proof. There are four contributions. First, the 44 paired boundary passages give the built-in Hom normalization $44(N - 1) = 44$ for $N = 2$; the raw closure removes it, contributing

$$d_1 = -44. \quad (60)$$

Second, the 227 separately labelled factors $X^{\otimes 227}$ each contribute +1 in the adopted convention:

$$d_2 = 227. \quad (61)$$

Third, the one-handle theorem contributes the sum of the 44 y -pairs and $271 = 44 + 227$ z -pairs:

$$d_3 = 44 + 271 = 315. \quad (62)$$

Finally the selected cabled state has $\alpha = 0$, $|r| = 2$, and $N = 2$, so the MWW cabled shift is

$$d_4 = (1 - N)(2|r| + |\alpha|) = (-1) \cdot 4 = -4. \quad (63)$$

Adding [equations \(60\)](#) to [\(63\)](#) gives

$$d_1 + d_2 + d_3 + d_4 = -44 + 227 + 315 - 4 = 494. \quad (64)$$

$\square$

The integer arithmetic in [equation \(64\)](#) is checked in Lean. Lean does not prove that these four integers are the correct geometric grading contributions for the actual filling; that binding is exactly why Assumption A2 remains visible.

10. DESCENT THROUGH THE ONE- AND TWO-HANDLE RELATIONS

10.1. The through-degree separation. Let F_d denote the span of endpoint diagrams of through degree at most d . Composition on either side cannot increase through degree. The selected cup U has through degree 86. The maps $\psi_i^{[0]}$ and $\psi_i^{[1]}$ also preserve through degree 86 and add the same one-cup factor. Thus neither through degree nor cup count distinguishes their images. Their statewise distinction in Proposition 10.3 comes from the core count of Lemma 10.2. The following elementary filtration lemma controls only relation terms that are independently known to lie in the low-cell subspace F_{84} .

Lemma 10.1 (Filtration argument). *Suppose P_{86} is a linear projector with image the through-86 top cell and kernel containing F_{84} . If $I \subset F_{84}$ is closed under left and right endpoint composition, then $P_{86}(I) = 0$. In particular, no vector in I can change $D_3(v_T)$.*

Proof. The inclusion $I \subset F_{84} \subset \ker P_{86}$ gives $P_{86}(I) = 0$. The last statement follows by linearity of every map after P_{86} in equation (42). $\square$

The point is not that a physical cap itself kills every lower cell. It need not. The projection is a linear postcomposition after the genuine quantum trace shadow. Its compatibility with the actual cabled relations is part of Assumption A3.

10.2. The local Frobenius calculation. For $b \geq 1$, let $\Delta^{(b-1)} : \mathbf{A} \rightarrow \mathbf{A}^{\otimes b}$ be the iterated coproduct, with $\Delta^{(0)} = \text{id}$. Put

$$D_b = \Delta^{(b-1)}(X) = X^{\otimes b}, \quad (65)$$

$$U_b = \Delta^{(b-1)}(1) = \sum_{a=0}^{b-1} X^{\otimes a} \otimes 1 \otimes X^{\otimes (b-1-a)}. \quad (66)$$

Lemma 10.2 (Counit identities). *For every $b \geq 1$,*

$$\epsilon^{\otimes b}(U_b) = 0, \quad \epsilon^{\otimes b}(D_b) = 1. \quad (67)$$

Proof. Formula (21) implies $\Delta^{(b-1)}(X) = X^{\otimes b}$ by induction. Applying $\epsilon^{\otimes b}$ gives $1^b = 1$. The corresponding induction for 1 gives the sum in equation (66); every summand contains exactly one factor 1, and $\epsilon(1) = 0$. Hence every summand evaluates to zero. $\square$

10.3. The cabled relation space. Let $\mathcal{C}_s$ be the raw summand at a finite cable state s and set $\mathcal{C}_{\text{raw}} = \bigoplus_s \mathcal{C}_s$. The two-handle relation space $\mathcal{R}_{2h}$ is spanned by

$$\beta_i(b)x - x, \quad \psi_i^{[0]}x, \quad \psi_i^{[1]}x - x \quad (68)$$

for every attaching component, state, admissible signed-colour-preserving braid, and source vector.

Proposition 10.3 (Cubic beta and psi equations). *Under Assumptions A2 and A3, the statewise rows $\Lambda_s^{(3)}$ satisfy*

$$\Lambda_s^{(3)}\beta_i(b) = \Lambda_s^{(3)}, \quad (69)$$

$$\Lambda_{s+e_i}^{(3)}\psi_i^{[0]} = 0, \quad \Lambda_{s+e_i}^{(3)}\psi_i^{[1]} = \Lambda_s^{(3)}. \quad (70)$$

Consequently $\mathcal{R}_{2h} \subseteq \ker \Lambda^{(3)}$.

Proof. At the base multiplicity the constant signed-permutation quotient is trivial and a pure residual acts as $I + O(h)$. Since $\rho_h(W) - I = O(h^3)$ on the whole endpoint module, postcomposition by a pure residual changes equation (42) only in order h^4 . This proves equation (69) at divided cubic order. Assumption A3 supplies a common physical-copy target and the same statement at every finite multiplicity.

For a forward pair addition, the two actual core counits act on the new Frobenius factors. Under the actual-to-canonical identification in Assumption A3, the undotted and once-dotted insertions are U_b and D_b . Lemma 10.2 gives respectively 0 and 1. The raw degree +2 of the once-dotted map cancels the target cabled shift -2 , so the second equation has total degree zero. This proves equation (70). Applying the three equations to the generators in equation (68) gives the kernel inclusion. $\square$

Corollary 10.4 (Two-handle descent). *There is a unique linear map*

$$\overline{\Lambda}_{2h}^{(3)} : \mathcal{C}_{\text{raw}}/\mathcal{R}_{2h} \longrightarrow \mathbb{Q} \quad (71)$$

whose pullback is $\Lambda^{(3)}$. On the selected class its value is 2624.

Proof. Proposition 10.3 places the relation space in the kernel, so the quotient universal property gives equation (71). The selected base row is the detector of definition 8.1; its value is equation (53). $\square$

11. THE CHOSEN SPHERES AND THREE-HANDLE DESCENT

11.1. The finite homology calculation. The public data supplies three rooted PL surface constructions. Their exact combinatorial sizes are

surface	leaves	pair-of-pants or split bands	root caps
TH1	350176	350175	1
TH2	229198	229197	1
THXY	11115	11114	1

For THXY the band count includes 312 macro bands and $5401 + 5401$ corridor split bands. The finite checker visits every stored row and verifies the rooted ancestry, stored normal-framing fields, terminal boundary reduction, SHA identities, schema, determinant, and verdict strings. Its geometric intersection test is narrower: only the comb bands have full coordinates for an all-row PL self-intersection and slab-gap calculation. The finger annuli, bigon, and cap have no such coordinates in the public data.

Exact computation 11.1 (Euler characteristics of the stored surfaces). Each of the three stored connected surfaces has Euler characteristic 2.

Proof. The rooted construction begins with one disk for each leaf, attaches one 1-handle for each band, and ends with one root cap. Therefore

$$\chi(\text{TH1}) = 350176 - 350175 + 1 = 2, \quad (72)$$

$$\chi(\text{TH2}) = 229198 - 229197 + 1 = 2, \quad (73)$$

$$\chi(\text{THXY}) = 11115 - 11114 + 1 = 2. \quad (74)$$

The ancestry tree is connected and the orientation data is coherent. Thus, within the supplied PL model, each surface is a genus-zero closed oriented surface. This calculation does not by itself place the model in the actual ∂W_2 ; that is Assumption A4. $\square$

The stored comb-band coordinate sectors are respectively

$$\text{THXY} \subset [2, 13/2], \quad \text{TH1} \subset [8, 9], \quad \text{TH2} \subset [10, 11]. \quad (75)$$

The interval separation proves the stated slab-gap property for the stored comb bands. It does not prove pairwise disjointness for the complete surfaces. Embedding the full models as disjoint surfaces in one actual post-two-handle boundary is Assumption A4.

Exact computation 11.2 (Unimodular class matrix). Let the three claimed sphere classes be the columns of

$$S = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}. \quad (76)$$

Then $\det S = 1$.

Proof. Expansion along the third row gives

$$\det S = (-1)(-40) + 1(-39) \quad (77)$$

$$= 40 - 39 = 1. \quad (78)$$

Here the first relevant minor is $(-189)(-269) - 41 \cdot 1241 = -40$, and the third is $(-1311)1241 - (-189)8608 = -39$, with the cofactor signs already included in equation (77). The same integer calculation is T73.sphereDet_eq_one in Lean. $\square$

Under Certified Fact C2 and Assumption A4, the columns of S are the classes of three pairwise-disjoint embedded spheres in the same ∂W_2 . External Result E7 then permits them to replace the actual attaching spheres up to the operations in External Result E7.

11.2. The six leading maps. For a chosen sphere j , let $b_j > 0$ be the number of newly created Frobenius tensor factors. Use the notation

$$D_j = X^{\otimes b_j}, \quad U_j = \sum_{a=0}^{b_j-1} X^{\otimes a} \otimes 1 \otimes X^{\otimes (b_j-1-a)}. \quad (79)$$

Let $Q_{s,0}$ and $Q_{t_j,0}$ be the constant terms of the full source and target coordinate transports. The abstract leading model is

$$F_{j,0}^{(0)} = Q_{t_j,0}^{-1}(\text{id}_{\text{old}} \otimes U_j)Q_{s,0}, \quad (80)$$

$$F_{j,1}^{(0)} = Q_{t_j,0}^{-1}(\text{id}_{\text{old}} \otimes D_j)Q_{s,0}. \quad (81)$$

Lemma 11.3 (Abstract sphere rows). *Let r be any row on the old factor and put $E_j = \epsilon^{\otimes b_j}$. For the maps in equations (80) and (81),*

$$(r \otimes E_j)F_{j,0}^{(0)} = 0, \quad (r \otimes E_j)F_{j,1}^{(0)} = r, \quad (82)$$

after using the displayed coordinate equivalences.

Proof. Coordinate conjugation cancels because the source and target rows are pulled back through the same $Q_{s,0}$ and $Q_{t_j,0}$. What remains is $E_j(U_j) = 0$ and $E_j(D_j) = 1$, which is Lemma 10.2. $\square$

Lemma 11.3 is an algebraic theorem. It does not identify the abstract maps with the MWW cobordism maps. That identification, on the whole source and for the actual chosen surfaces, is exactly Assumption A4.

Proposition 11.4 (Three-handle kernel equations). *Under Assumption A4, the descended two-handle row ℓ_1 satisfies, for each $j \in \{\text{TH1}, \text{TH2}, \text{THXY}\}$,*

$$\ell_1 \circ \Sigma_{j,0} = 0, \quad \ell_1 \circ (\Sigma_{j,1} - \text{id}) = 0. \quad (83)$$

Proof. Assumption A4 gives the whole-source naturality square identifying the actual constant terms of $\Sigma_{j,0}$ and $\Sigma_{j,1}$ with equations (80) and (81). The old divided row begins in order h^3 ; the same assumption places every positive-order transport correction in order at least h^4 after composition. Hence the divided cubic depends only on the constant maps. Lemma 11.3 then yields the two equations. $\square$

12. THE COMPLETE QUOTIENT AND THE CONDITIONAL CONCLUSION

Let W_0 be the one-handle coefficient trace, W_1 its quotient by all two-handle relations, and W_2 the quotient of W_1 by the six chosen-sphere relations. Let $q_{01} : W_0 \rightarrow W_1$ and $q_{12} : W_1 \rightarrow W_2$ be the quotient maps. Denote the selected class in W_0 by x_0 .

Lemma 12.1 (A linear detector survives a quotient). *Let V be a vector space, $R \subseteq V$ a subspace, and $\lambda : V \rightarrow \mathbb{Q}$ a linear map with $R \subseteq \ker \lambda$. Then there is a unique $\bar{\lambda} : V/R \rightarrow \mathbb{Q}$ such that $\bar{\lambda} \circ \pi = \lambda$. If $\lambda(x) \neq 0$, then $\pi(x) \neq 0$.*

Proof. Define $\bar{\lambda}(x + R) = \lambda(x)$. If $x - y \in R$, then $\lambda(x) - \lambda(y) = 0$, so the definition is independent of the representative. Uniqueness follows from surjectivity of π . Finally, $\pi(x) = 0$ would imply $x \in R$ and hence $\lambda(x) = 0$. $\square$

Proposition 12.2 (Nonzero class after all relation quotients). *Under Assumptions A2–A4, there is a linear functional $\ell_2 : W_2 \rightarrow \mathbb{Q}$ such that*

$$\ell_2(q_{12}q_{01}x_0) = 2624. \quad (84)$$

In particular, $q_{12}q_{01}x_0 \neq 0$.

Proof. Lemma 6.3 first descends the quantum shadow detector through the coefficient trace. Proposition 10.3 and Lemma 12.1 descend it through q_{01} . Proposition 11.4 and the same quotient lemma descend it through q_{12} . At every stage the pullback agrees with the preceding row, so equation (53) gives equation (84). Nonvanishing follows from the last sentence of Lemma 12.1. $\square$

Proposition 12.3 (Closed candidate class). *Under Assumptions A4 and A5, the class in proposition 12.2 maps to a nonzero class*

$$z_{494} \in \mathcal{S}_{0,494}^2(X(41, 189, 73); \mathbb{Q}). \quad (85)$$

Proof. Assumption A4, through External Result E4, identifies W_2 with the post-three-handle module after exactly the six displayed relations. Assumption A5, through External Result E5, supplies a grading-preserving isomorphism after the four-handle attachment. Isomorphisms preserve nonzero elements, and Proposition 9.1 supplies the quantum degree. $\square$

Proof of Theorem 1.1. Proposition 12.3 gives a nonzero element in quantum degree 494. By Assumption A5, the same graded piece for S^4 is zero. If the candidate were diffeomorphic to S^4 , the same assumption, together with External Result E6, would give a grading-preserving linear isomorphism between these two pieces, which is impossible. Thus the candidate is not diffeomorphic to S^4 . Corollary 4.5 supplies the homotopy-sphere conclusion. $\square$

Remark 12.4 (The S^4 control is a zero-to-zero control). When the point-push word is replaced by the identity, the operator $\rho_h(W) - I$ vanishes and the endpoint detector sends the standard control to zero. The published S^4 module is also zero in quantum degree 494. Thus this control checks compatibility of two zeros; it is not a positive calibration that independently demonstrates the geometric binding of the candidate detector.

Remark 12.5 (Presentation and serialization dependence). The raw coefficient $[h^3]\ell(\rho_h(W) - I)u$ is a quantity of a based, labelled, oriented movie presentation. Reordering the same crossing rows by the emitter's increasing-coordinate order produces a different series. With the corrected collar endpoints its cubic coefficient is 0 and its first nonzero coefficient is $663872h^4$. That order reverses 168 non-disjoint events and is not a legal serialization under Assumption A1. The former value -58976 used the same mixed endpoint coordinates as the superseded public value. The calculation nevertheless shows why presentation independence cannot be inferred from the scalar alone: it must be proved through simultaneous transport of every part of the naturality square.

Remark 12.6 (Insertion-point dependence). The point-push operator and cap row are inserted at a specified collar and a specified point. Moving the insertion point is harmless only if the move is accompanied by the corresponding trace/cyclicity and collar-isotopy square. That geometric statement is Assumption A1. The exact word calculation does not prove it.

13. A FULLY WORKED FINITE-DIMENSIONAL EXAMPLE

This section illustrates the entire algebraic mechanism on a small example. It is not a four-manifold calculation and supplies no evidence for any candidate-specific assumption. Its purpose is to make the order of operations—word action, coefficient extraction, relation killing, quotient, and nonvanishing—checkable by hand.

Let $R_0 = \mathbb{Q}[\varepsilon]/(\varepsilon^4)$, $t = 1 + \varepsilon$, and $E_0 = R_0^3$ with basis e_0, e_1, e_2 . The two positive Burau generators are

$$B_1 = \begin{pmatrix} -\varepsilon & 1 + \varepsilon & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\varepsilon & 1 + \varepsilon \\ 0 & 1 & 0 \end{pmatrix}, \quad (86)$$

$$B_1^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ t^{-1} & 1 - t^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & t^{-1} & 1 - t^{-1} \end{pmatrix}, \quad (87)$$

where $t^{-1} = 1 - \varepsilon + \varepsilon^2 - \varepsilon^3$.

Take the pure commutator word

$$w_0 = \sigma_1^2 \sigma_2^2 \sigma_1^{-2} \sigma_2^{-2}. \quad (88)$$

Multiplying the eight matrices from left to right and reducing modulo ε^4 gives

$$\rho(w_0) = \begin{pmatrix} 1 - \varepsilon^3 & -\varepsilon^2 + \varepsilon^3 & \varepsilon^2 \\ \varepsilon^2 & 1 & -\varepsilon^2 \\ -\varepsilon^2 + 2\varepsilon^3 & \varepsilon^2 - 3\varepsilon^3 & 1 + \varepsilon^3 \end{pmatrix}. \quad (89)$$

The equality can be checked by multiplying $B_1 B_1 B_2 B_2 B_1^{-1} B_1^{-1} B_2^{-1} B_2^{-1}$ using [equations \(86\) and \(87\)](#); no omitted matrix is involved.

Let $u_0 = e_0 - e_1$ and let $\ell_0 = e_0^*$. Define the toy detector by the same pattern as [definition 8.1](#), but at order two:

$$d_2(v) = [\varepsilon^2] \ell_0(\rho(w_0) - I)v. \quad (90)$$

Reading the coefficient of ε^2 from the first row of [equation \(89\)](#) gives the complete detector matrix

$$d_2 = \begin{pmatrix} 0 & -1 & 1 \end{pmatrix}. \quad (91)$$

Therefore

$$d_2(u_0) = \begin{pmatrix} 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 1. \quad (92)$$

Now model two relation maps by the columns

$$r_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad r_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}. \quad (93)$$

The matrix product

$$d_2 R = \begin{pmatrix} 0 & 0 \end{pmatrix} \quad (94)$$

is the complete kernel check. Hence d_2 descends to $E_0/\text{im } R$. A concrete quotient coordinate is

$$q : E_0 \longrightarrow R_0, \quad q(x_0, x_1, x_2) = -x_1 + x_2, \quad (95)$$

whose matrix is the same row as [equation \(91\)](#). The relations vanish: $q(r_0) = q(r_1) = 0$, while $q(u_0) = 1$. Thus the selected class is nonzero in the quotient.

This toy example contains the exact logical skeleton of the candidate argument. What it lacks is precisely the geometric content: there is no claim that the toy relation columns arise from handle maps or that its final quotient is a smooth invariant.

14. FORMALIZATION BOUNDARY

The Lean development has two jobs. First, it checks the finite arithmetic and the algebra of linear quotients. Second, it fixes the exact type of the conditional implication so that a hidden strengthening of the conclusion or weakening of the premises is difficult. It does not define smooth four-manifolds, skein lasagna modules, actual handle movies, or the geometric predicate that a stored surface is an attaching sphere.

14.1. What is kernel-checked. The following statements are proved from definitions in Lean.

- (1) The two determinant identities in [exact computation 4.2](#), the chosen sphere determinant in [exact computation 11.2](#), the integer identity $(-2)^3 \cdot (-328) = 2624$, and the grading sum $-44 + 227 + 315 - 4 = 494$.
- (2) The coefficient- HH_0 quotient, the fact that a cyclic shadow kills its relation span, and the universal linear factorization through the quotient.
- (3) The Frobenius counit identities $\epsilon^{\otimes b}(U_b) = 0$ and $\epsilon^{\otimes b}(D_b) = 1$, including arbitrary positive split trees and coordinate conjugation.
- (4) The fact that six assumed sphere-map bindings place their relation ranges in the kernel of the detector, and the induced nonzero-class statement in the quotient.
- (5) The implication from the stated external interfaces to “homotopy sphere and not diffeomorphic to S^4 .”

The 45,360-letter matrix product is not executed by Lean. It is executed by the independent exact Python program described in [section 8](#). Lean imports the final primitive integer -328 for the short substitution check. This is why the Python computation and the Lean implication are complementary rather than redundant.

14.2. **What is stored as a field.** The core Lean theorem has the schematic form

$$\text{NotStandardInterfaces} \longrightarrow \neg \text{Diffeomorphic}(\text{candidate}, S4). \quad (96)$$

The interface contains the actual- HH_0 binding, the actual detector row, the beta/psi maps, the three actual sphere maps, the MWW coequalizer and transport, the four-handle map, standard-sphere support, and diffeomorphism invariance. Supplying a field of a structure is an assumption, not a proof of the field's geometric proposition. In particular, compilation of [equation \(96\)](#) proves only that these fields imply the conclusion.

Table 1: External theorem, paper item, and Lean-field crosswalk.

External result	paper item	principal Lean field or definition
MWW Theorem 4.7	E1; used in A2	<code>OneHandleInterface.hh0Binding</code> , <code>hh0Lift</code> , <code>hh0LiftCommutes</code> , <code>hh0LiftUnique</code>
BHPW Theorem 4.6	E9; used in A2	candidate-side input to <code>OneHandleInterface.rawCapBinding</code> and the naturality fields of <code>SphereChart</code>
BPW Proposition 3.20; Theorem 3.21	E8; used in A2	<code>OneHandleInterface.rawCap</code> , <code>rawCapKillsHH0</code> ; algebraic quotient in <code>CoefficientTrace.lean</code>
MWW Definition 3.1; Theorem 3.2	E2; used in A3	<code>BetaPsiInterface.betaRelationBinding</code> , <code>psi0Binding</code> , <code>psi1Binding</code> , <code>q01</code>
Horvat–Jabłonowski Theorem 5.3	E7; used in A4	<code>SphereMWWFamily.hjBinding</code> and <code>SphereTriple.hjBinding</code>
MWW Theorem 3.7; Theorem 3.10	E3–E4; used in A4	<code>SphereMWWFamily.mwCoequalizerBinding</code> , <code>q12</code> , <code>mwwTransport</code>
MWW Proposition 3.4 and Corollary 3.5	E5; used in A5	<code>FourHandleInterface.fourIso</code> , <code>fourHandleBinding</code>
MWW Definition 5.4 and Example 5.6	E6; used in A5	<code>S4ControlInterface.degreeSupport</code> ; alternatively <code>S4ReductionData.evalB4</code> and <code>attach4</code>
Iwaki Proposition 2.1	E10; used in A1	<code>CappellShanesonInterface.matrixConditionsToHomotopySphere</code>
MWW smooth invariance	E6; used in A5	<code>DiffeomorphismInvarianceInterface.preservesGradedObject</code>
van Kampen theorem	E11 (alternative route)	<code>CSTopologyData.vanKampen</code>
Wang and Mayer–Vietoris sequences	E12 (alternative route)	<code>CSTopologyData.wangMayerVietorisPoincare</code>
Poincaré duality	E13 (alternative route)	<code>CSTopologyData.wangMayerVietorisPoincare</code>
Hurewicz and Whitehead theorems	E14 (alternative route)	<code>CSTopologyData.hurewiczWhitehead</code>

14.3. **Compilation and axioms.** The recorded clean replay used Lean 4.32.1 and mathlib commit 520045ab14e26149ee970e2e617ca04b09bde5d6. It compiled every audited project module in a fresh output root and then ran `lakeleanT73Audit.lean`. The file requests `#print axioms` for 38 declarations. The recorded partition was two reports using only `propext`, five using `propext`, `Quot.sound`, and thirty-one using `propext`, `Classical.choice`, `Quot.sound`. No report contained `sorryAx`.

These are foundational dependencies from mathlib and quotient constructions; they do not certify the geometric interfaces. A reader can reproduce both the compilation and the axiom scan using the commands in [section 15](#).

15. REPRODUCTION

The public repository is

<https://github.com/toffee-desuwa/smooth4pc-t73-lean>.

This erratum revision is identified by the repository tag `v0.2-erratum1`. The commands below assume a fresh clone and Python 3.10 or later.

15.1. Reconstruct the detector. Run

```
python -I -B scripts/recompute_t73_delta3.py --check
```

The expected bearing lines are

```
INPUT_SHA256=0F0F4A568D966367313C29C25CEDDE4FD9ABAD32B9375519E9CDF9168AE0A764
B44_LENGTH=11340
B44_SHA256=7C2D2F792C2672221A76CAF08A71F560AF0CB7654B4D537C00FAD00B16EFA187
B88_LENGTH=45360
B88_SHA256=C13D0A3BA9B05F41A6D2C5B4AB12DDECDABEFC1883835891AD3BD2B8955B5FFB
POSITION_TABLE_SHA256=119C7E9E74AE6C820DA72A84CDFD5D445D81E6C3AACC209C25D37C323961508
ELL_RHOW_MINUS_I_U_EPS=[0,0,0,-328,14596,-410246,9595271]
ELL_RHOW_MINUS_I_SQUARED_U_EPS=[0,0,0,0,0,-102729600]
DELTA3_ETA_T1=2624
DELTA3_XI=0
VERIFY=PASS
```

The script itself has SHA-256 27B6D1E11CF60FFD13E041B9836A36178EBDA62322FFB046523D55EB67F45CFD. The canonical JSON receipt produced from the current input has SHA-256 7C681B99492A5C34B3B8A7EC8B7269E3FD67CD989728C2D4A1C619A63387A722.

15.2. Verify the public geometric data. Run

```
python -I -B scripts/verify_public_geometry_evidence.py
```

This command checks the 29 bundled files, their SHA-256 identities, the chosen-sphere determinant, and the global-descending calculation from the full 2,126,291-crossing PD input. Passing this command certifies only the finite stored model. It does not prove Assumptions [A1](#), [A4](#), or [A4](#).

15.3. Compile and inspect the Lean development. After installing elan, run

```
lake update
git diff --exit-code -- lake-manifest.json
lake exe cache get
python -B tests/test_t73_minimal_formalization.py -v
lake lean T73Audit.lean
```

The Python test requires two passing tests, compiles the audited modules into a fresh temporary output root, requires exactly 38 axiom reports, rejects any axiom outside `propext`, `Classical.choice`, and `Quot.sound`, and rejects `sorryAx`.

15.4. Build this paper. On Windows, with Tectonic 0.17.0 installed at the path recorded by `paper/build.ps1`, run

```
powershell -ExecutionPolicy Bypass -File paper/build.ps1
```

On Linux or macOS with `tectonic` on `PATH`, the cross-platform one-line command is

```
cd paper && tectonic t73_candidate.tex --outdir build --keep-logs
```

The Windows binary used for this draft is Tectonic 0.17.0 with SHA-256 99FFCFDBF1EBF8BDDA9E791942E3D06AEDB12463FDDC33F07DE6F5211C8BF08D.

16. REVIEW BOUNDARY

The finite computation is an ordinary, reproducible matrix calculation. Its nonzero value is not the principal uncertainty. The principal uncertainty is whether the stated coefficient, collar, cabled state, and sphere maps are the actual geometric maps to which the published quotient theorems apply. The most direct review strategy is therefore not to rerun the last integer multiplication first. It is to attack Assumptions `A1`–`A5` as geometric statements, in particular the whole-source sphere-map binding in `A4`.

Remark 16.1 (No inference from filenames or certificates). Names such as “global descending certificate”, “chosen sphere receipt”, or “Hattori decision” are locators for data. They do not occur as premises of any proof in this article. Whenever a finite file is used, the relevant mathematical content is stated explicitly and the replay command is given. Whenever a geometric conclusion is not derived from those finite data, it is an Assumption.

Remark 16.2 (The legality audit is bounded). The pre-search legality criteria were committed at `cf4a990` before the detailed adjudication at `9d75dcd`. Of the 22 tested alternative standard-member collars, none was proved legal: 19 were rejected and 3 remained unestablished. This does not prove that every legal standard-member collar has zero detector. It shows only that the registered `t73` chronology survives that bounded test.

17. HOW THE FIVE ASSUMPTIONS CAN BE DISCHARGED OR REFUTED

The five assumptions are not generic requests for “more justification.” Each has a definite proof object and a definite failure mode. This section states both so that independent review can be adversarial rather than impressionistic.

17.1. A1: presentation and collar. To discharge A1, one must give a framed Kirby movie from the standard Cappell–Shaneson surgery construction for [equation \(2\)](#) to the frozen cut presentation. The movie must transport every attaching-component label, normal framing, basepoint, collar coordinate, and the detector insertion disk. At each interchange of events, the two supports must be disjoint or an explicit movie move must be cited. The final based labelled point-push word must be the registered word generated in [definitions 7.4](#) and [7.5](#).

A1 is refuted by any forced change of framing, basepoint, insertion point, or event order that is not accompanied by the simultaneous coordinate transport of [equation \(99\)](#). It is also refuted if two legal movies for the same fixed geometric input give different scalar series. The existing legality audit rejects the known emitter ordering, but it is not a general uniqueness theorem for all legal collars.

17.2. A2: Hattori input and endpoint model. To discharge A2, one must construct the actual coefficient equivalence [equation \(23\)](#) from the paired annuli and prove both action-naturality squares. Applying the quantum trace and the 227 counits must then give the specific vector $u = e_2 - e_{87}$, with the covector $\ell = e_{87}^* - e_2^*$ arising from the same collar position table. The proof must also bind the actual word to the third pure-braid filtration term on the whole endpoint module and derive the four grading contributions from that same filling.

A2 is refuted if the actual diagonal class maps to a different endpoint vector, if the cap row changes without simultaneous conjugation, if the word has a lower-order term on any endpoint vector, or if one grading contribution belongs to a different normalization. Agreement of the single scalar 2624 is not enough to rule out these failures.

17.3. A3: the all-state two-handle cocone. To discharge A3, one must define the statewise rows for every finite cable state and prove [equations \(69\)](#) and [\(70\)](#) as equalities of linear maps on their full source spaces. The common-target and sign choices must be compatible around every square of the cable-state graph. A proof on the selected vector, a generic-rank computation, or a finite list of cable states does not establish A3.

A3 is refuted by a single admissible braid or stabilization source vector on which one of those map equalities fails. Such a counterexample would show that the detector does not descend through the MWW two-handle quotient even though its value on the base representative is nonzero.

17.4. A4: actual spheres and whole-source sphere maps. To discharge A4, the finite surfaces of $C2$ must first be embedded in one actual ∂W_2 with their framings, disjointness, and homology classes preserved. The hypotheses of External Result E7 must then be verified for that same handle decomposition. Finally, the two MWW cobordism maps for each sphere must be identified, on the whole source, with the conjugated leading maps [equations \(80\)](#) and [\(81\)](#); their positive-order transport terms must be shown not to enter the divided cubic.

A4 is refuted if any chosen surface fails to bind to the actual boundary, if the replacement theorem’s boundary decomposition or basis hypothesis fails, or if one actual sphere-map vector has a nonzero detector

value contrary to [equation \(83\)](#). Because this is the last relation layer before the closed invariant, A4 is the most consequential unresolved point.

17.5. A5: rational coefficients and grading. To discharge A5, one must trace the coefficient functor from the integral B^4 evaluation and S^4 computation to the rational module used here, showing that tensoring introduces neither a Tor correction nor a grading change. The four-handle isomorphism must be checked to have bidegree $(0, 0)$, and the diffeomorphism-induced equivalence must preserve the same quantum grading.

A5 is refuted if the rational degree-494 piece of S^4 is not zero in the chosen convention, or if either comparison map shifts degree. Unlike A3 and A4, this question is largely a matter of matching published coefficient and grading conventions, so it should be settled before undertaking further large finite computations.

APPENDIX A. THE COMPLETE DEGREE-THREE ENDPOINT VECTOR

With the conventions of [section 7](#), the coefficient vector in [equation \(49\)](#) is

$$(a_0, \dots, a_{87}) = (-7052, -7052, 164, 164, 332, 332, 324, 324, 316, 316, 308, 308, \\ 300, 300, 292, 292, 284, 284, 276, 276, 268, 268, 260, 260, \\ 252, 252, 244, 244, 236, 236, 228, 228, 220, 220, 212, 212, \\ 204, 204, 196, 196, 188, 188, 180, 180, 172, 172, 164, 164, \\ 156, 156, 148, 148, 140, 140, 132, 132, 124, 124, 116, 116, \\ 108, 108, 100, 100, 92, 92, 84, 84, 76, 76, 68, 68, \\ 60, 60, 52, 52, 44, 44, 36, 36, 28, 28, 20, 20, \\ 12, 12, -164, -164). \quad (97)$$

The detector covector is $\ell = e_{87}^* - e_2^*$, so the contraction is visibly

$$\ell(a_0, \dots, a_{87})^T = a_{87} - a_2 = -164 - 164 = -328. \quad (98)$$

The sum of the 88 entries is zero, as expected for the unreduced Burau difference on the invariant total-coordinate functional; this is a check, not an input to [equation \(98\)](#).

APPENDIX B. THE FOUR GRADING CONTRIBUTIONS

For ease of audit, the four contributions of [section 9](#) are collected without abbreviation:

term	source	calculation
-44	remove the $p(N-1)$ Hom normalization for $p = 44, N = 2$	$-44(2-1) = -44$
+227	227 distinct Frobenius labels, each of degree +1	$227 \cdot 1 = 227$
+315	one-handle gluing shift from 44 y -pairs and 271 = 44 + 227 z -pairs	$44 + 271 = 315$
-4	cabled shift for $N = 2, \alpha = 0, r = 2$	$(1-2)(2 \cdot 2 + 0) = -4$
total		$-44 + 227 + 315 - 4 = 494$

APPENDIX C. CONVENTION-LEGALITY CRITERIA

A serialization is legal only if it describes the same framed handle presentation, the same oriented based cut tangle, the same labelled boundary pairing, the same normal field, and the same insertion point. Its event order must be a linear extension of the geometric partial order. Adjacent events may be exchanged only when their supports are disjoint and a strict interchange theorem applies. Word replacement requires equality in the based labelled braid or tangle category; equality of closures, permutations, free-group words, or homology classes is insufficient.

A coordinate change P must act simultaneously:

$$W' = PWP^{-1}, \quad u' = Pu, \quad \ell' = \ell P^{-1}, \quad (99)$$

with the shadow, collar, basepoint, and insertion point transported in the same naturality square. Under equation (99),

$$\ell'(W' - I)u' = \ell P^{-1}P(W - I)P^{-1}Pu = \ell(W - I)u. \quad (100)$$

Thus coordinate invariance is proved when the simultaneous transport exists; it is not imposed by selecting the preferred numerical answer.

A collar is legal only if it is an actual framed tubular collar of the fixed component and is isotopic to the registered collar relative to the boundary, basepoints, labels, old input balls, normal framing, and insertion point. A nontrivial relative mapping-class element or an added framing twist changes the input. These criteria were written before the detailed alternative collar outputs were opened. Their use in the candidate geometry remains Assumption A1.

APPENDIX D. TERMINOLOGY

Actual map: A map induced by a specified geometric tangle or surface cobordism, as opposed to an abstract linear map having the desired source and target.

Candidate-specific binding: An identification between immutable finite data and the actual geometric object required by a published theorem.

Complete quotient: The quotient by all relation generators in the applicable MWW handle formula, not by a sampled finite subset.

Divided cubic: The reduction modulo h of a map known on its whole domain to be divisible by h^3 , as in equation (43).

Endpoint shadow: The image of a coefficient trace class under the quantum vertical-to-horizontal trace and the fixed endpoint functor.

Finite certificate: Immutable finite data plus a deterministic exact checker. The term has no geometric force beyond the predicates actually recomputed.

Registered movie order: The oriented point-push chronology fixed in the public primitive input. In standard mathematical language it is the chosen linear extension of the geometric event partial order.

Material factor: In the earlier project notes this meant a newly created Frobenius tensor factor. The paper uses the latter standard phrase.

Reynolds average: The finite group average over physical-copy permutations. It is used only in the assumed all-state beta/psi construction, not as an independent geometric theorem.

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